GenAI Essentials Supplement

FOUNDATIONAL MODELS

A foundation model is trained on a large dataset to learn general patterns and features from the data. To read the same amount of text that was used to train GPT3 for example the average human would need to read 24X7 for the next 2600 years. Larger models since then were trained on much more.

Released in 2018, Bidirectional Encoder Representations from Transformers (BERT) was one of the first foundation models. BERT is a bidirectional model that analyzes the context of a complete sequence then makes a prediction. It was trained on a plain text corpus and Wikipedia using 3.3 billion tokens (words) and 340 million parameters. BERT can answer questions, predict sentences, and translate texts.

GPT-3 has a 96-layer neural network and 175 billion parameters and is trained using the 500-billion-word Common Crawl dataset. The popular ChatGPT chatbot is based on GPT-3.5. And GPT-4, the latest version, launched in late 2022 and successfully passed the Uniform Bar Examination with a score of 297 (76%).

Prompt Engineering

COSTAR

- Context – Providing background information helps the FM understand the specific scenario and provide relevant responses
- Objective – Clearly defining the task directs the FM's focus to meet that specific goal
- Style – Specifying the desired writing style, such as emulating a famous personality or professional expert, guides the FM to align its response with your needs
- Tone – Setting the tone makes sure the response resonates with the required sentiment, whether it be formal, humorous, or empathetic
- Audience – Identifying the intended audience tailors the FM's response to be appropriate and understandable for specific groups, such as experts or beginners
- Response – Providing the response format, like a list or JSON, makes sure the FM outputs in the required structure for downstream tasks

https://aws.amazon.com/blogs/machine-learning/implementing-advanced-prompt-engineering-with-amazon-bedrock/

ANTHROPIC PROMPT ENGINEERING BEST PRACTICES

META PROMPT ENGINEERING BEST PRACTICES

https://www.promptingguide.ai/introduction/tips

OUTPUT FORMATING

--output format
Please extract the name, size, price, and color from this product description and output it within a JSON object.

<description>The SmartHome Mini is a compact smart home assistant available in black or white for only $49.99. At just 5 inches wide, it lets you control lights, thermostats, and other connected devices via voice or app—no matter where you place it in your home. This affordable little hub brings convenient hands-free control to your smart devices.</description>

CHAIN OF THOUGHT

Are both the directors of Jaws and Casino Royale from the same country? Think step by step.

TREE OF THOUGHT PROMPT

- Bob is in the living room.**-** He walks to the kitchen, carrying a cup.**-** He puts a ball in the cup and carries the cup to the bedroom.**-** He turns the cup upside down, then walks to the garden.**-** He puts the cup down in the garden, then walks to the garage.**Question**: Where is the ball?
Self-consistency is like asking the same expert to solve a problem multiple times using different approaches, then taking the most common answer. Self-consistency is valuable whenever:

- The task has potential for reasoning errors
- Multiple valid approaches exist
- The problem benefits from diverse perspectives
- You need increased confidence in the result

The technique is about finding robust answers through consensus across different reasoning paths, which applies to virtually any complex problem.

```
"If a store offers a 20% discount on a $80 item and then applies
a 10% coupon on the discounted price, what is the final price?"
```

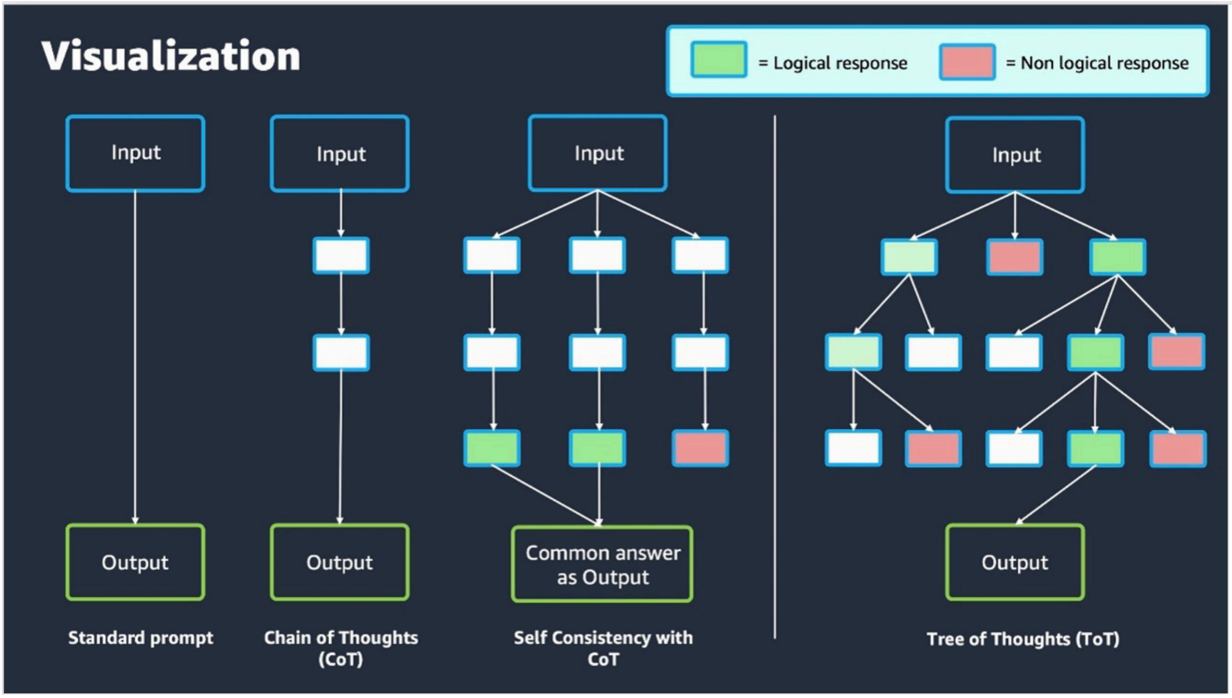
PII REMOVAL — DETAILED INSTRUCTIONS

Please follow these steps:

1. Replace all instances of names, phone numbers, card numbers, and home and email addresses with 'XXX'.
2. If the text contains no PII, copy it word-for-word without replacing anything.
3. Output only the processed text, without any additional commentary.

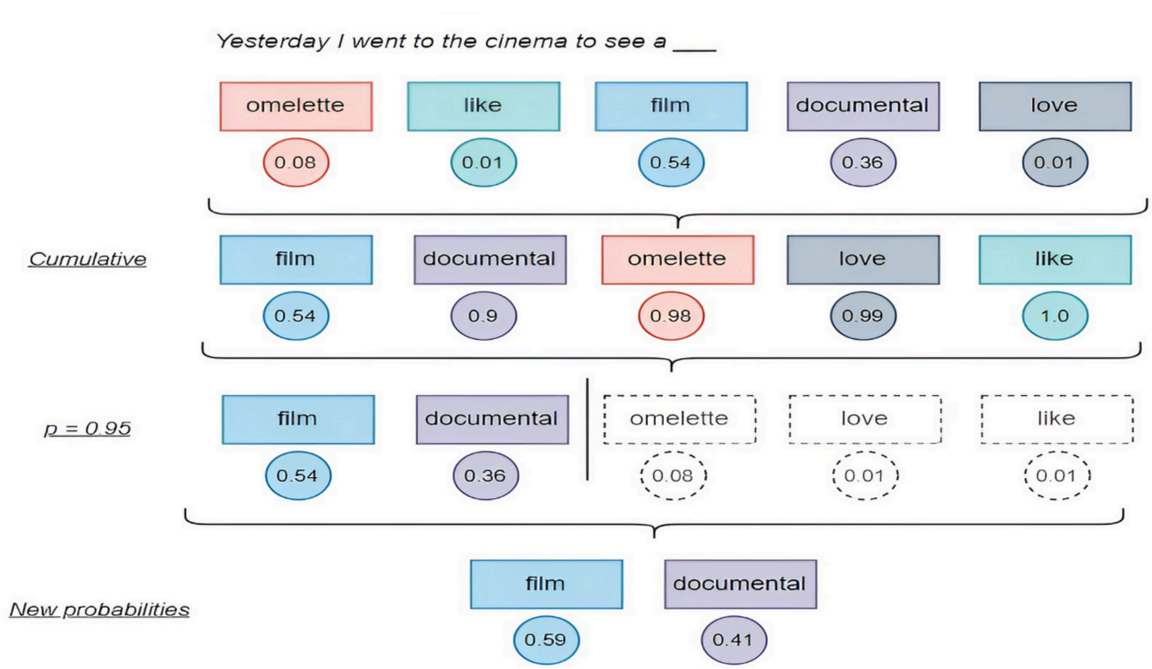
Here is the text to process:
Customer feedback for Sunshine Spa, 123 Main St, Anywhere. Send comments to Alice at sunspa@mail.com.
I enjoyed visiting the spa. It was very comfortable but it was also very expensive. The amenities were ok but the service made the spa a great experience.

Comparing prompts visually

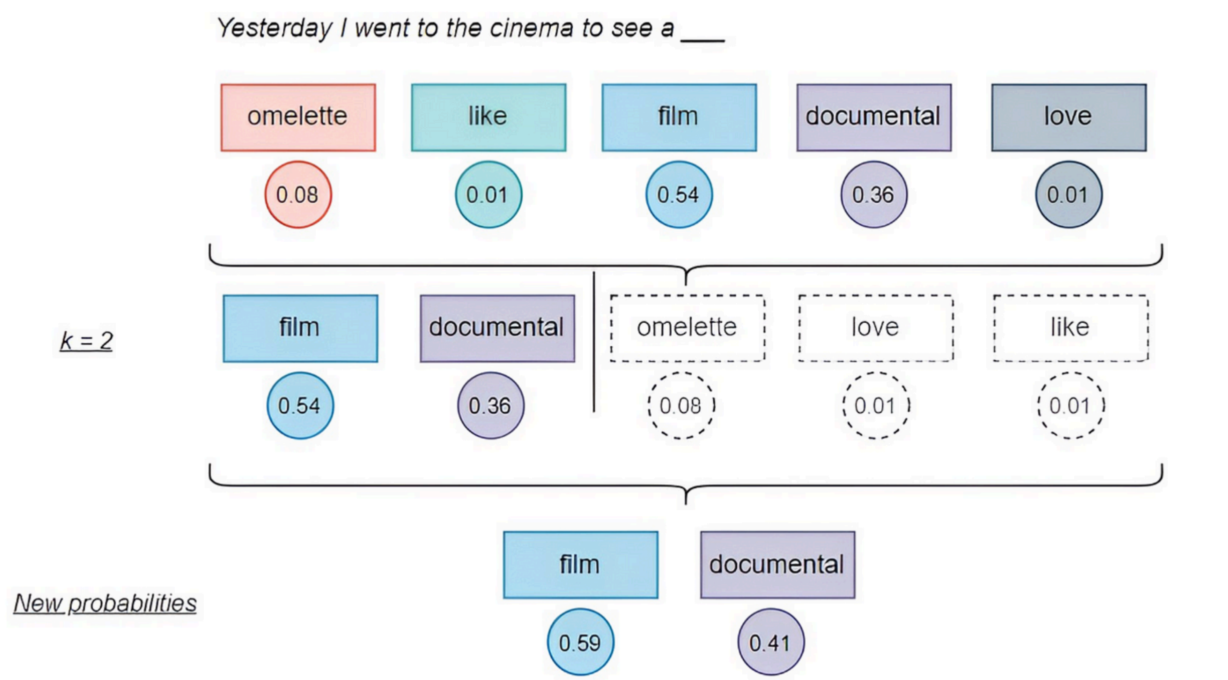


https://medium.com/@daniel.puenteviejo/the-science-of-control-how-temperature-top-p-and-top-k-shape-large-language-models-853cb0480dae

INFERENCE PARAMETERS TOP P



INFERENCE PARAMETERS TOP K



Responsible AI

While we can't directly audit the internal algorithms or training data of foundation models, these practical approaches can help identify, measure, and mitigate biases in their application within your specific context.
https://bhm.scholasticahq.com/article/38021-misguided-artificial-intelligence-how-racial-bias-is-built-into-clinical-models

PRACTICAL APPROACHES TO MITIGATE BIAS IN FOUNDATION MODELS

1. PROMPT ENGINEERING AND TESTING

- **Systematic prompt testing**: Test models with diverse scenarios and subjects to identify biased responses
- **Counterfactual testing**: Change attributes like gender, race, or age in prompts to detect differential treatment
- **Adversarial prompting**: Deliberately probe for biased responses to understand model limitations

2. FINE-TUNING AND ALIGNMENT

- **Domain-specific adaptation**: When possible, fine-tune models on carefully curated datasets representing diverse perspectives
- **RLHF (Reinforcement Learning from Human Feedback)**: Use diverse human evaluators to guide model behavior toward fairness
- **Parameter-efficient tuning**: Methods like LoRA allow targeted adjustments without full retraining

3. OUTPUT FILTERING AND POST-PROCESSING

- **Content moderation layers**: Implement filters to catch potentially biased or harmful outputs

- **Fairness evaluation metrics**: Apply metrics to model outputs to quantify bias across different demographic groups
- **Human-in-the-loop review**: Establish processes for human review of model outputs in sensitive contexts

4. MODEL SELECTION AND EVALUATION

- **Model cards**: Review documentation about known limitations and biases before selecting a foundation model
- **Comparative testing**: Evaluate multiple models on the same tasks to identify those with fewer biases
- **Red-teaming**: Employ specialized teams to systematically probe for biases and vulnerabilities

5. TRANSPARENCY AND DOCUMENTATION

- **Document limitations**: Clearly communicate known biases and limitations to users
- **Usage guidelines**: Develop clear guidance on appropriate and inappropriate use cases
- **Feedback mechanisms**: Create channels for users to report biased outputs

NOVA MODEL CARDS

https://docs.aws.amazon.com/ai/responsible-ai/nova-micro-lite-pro/overview.html

Goverance

Guidelines from NIST AI Risk Management Framework
https://www.nist.gov/itl/ai-risk-management-framework

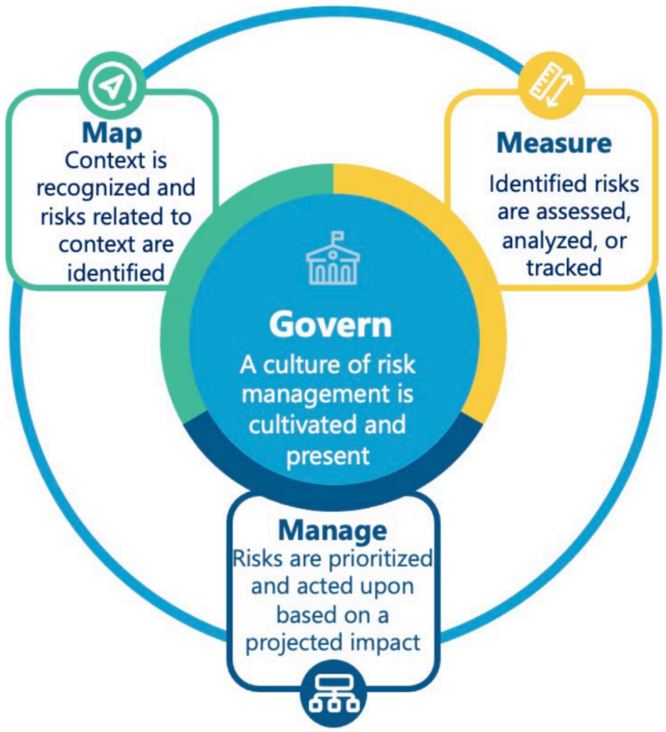


Figure 5: Functions organize AI risk management activities at their highest level to govern, map, measure, and manage AI risks. Governance is a cross-cutting function that is infused throughout and informs the other functions of the process.

GOVERN

Governance is designed to ensure risks and potential impacts are identified, measured, and managed effectively and consistently. Governance focuses on technical aspects of AI system design and development as well as on organizational practices and competencies that directly affect the individuals involved in training, deploying, and monitoring of such systems. Governance should address supply chains,including third-party software or hardware systems and data as well as internally developed AI systems.

MAP

The Map function establishes the context to frame risks related to an AI system. The information gathered while carrying out this function enables risk prevention and informs decisions for processes such as model management, and an initial decision about appropriateness or the need for an AI solution.

MEASURE

The Measure function employs quantitative, qualitative, or mixed-method tools, techniques, and methodologies to analyze, assess, benchmark, and monitor AI risk and related impacts.

MANAGE

The Manage function entails allocating risk management resources to mapped and measured risks on a regular basis and as defined by the Govern function. ex decrease the likelihood of system failures and negative impacts, increase transparency and accountability

Implementing GenAI Projects

GLUE

The General Langage Understanding Evaluation (GLUE) benchmark contains nine different tasks for evaluating fundamental language abilities. For example, one key task is textual entailment this involves is one sentence logically implies another sentence.
Sentence 1: The woman walked her dog through the park.
Sentence 2: (hypothesis) A person was moving a canine on a leash in a public garden area.
The model must determine if the hypothesis is describing the same scenario.

SUPERGLUE

Builds on GLUE on key test it provides is pronoun coreference resolution.
Sentence: Mary walked her dog. She enjoyed the fresh air. Model must determine that “She” refers to Mary and not the dog.

EVALUATION METRICS BLEU

Bilingual Evaluation Understudy measures how similar an AI-generated translation is to human reference translation. It counts matching words and phrases. This assesses if AI translation captures the same meaning as the original.

ROUGE

Recall-Oriented Understudy for Gisting Evaluation compares machine generated summaries to human summaries. It looks for overlapping content words and phrases. Higher ROGUE scores indicate summaries that better capture key information.

BERTSCORE

BERTScore is a neural evaluation metric designed specifically to address the limitations of traditional lexical metrics like BLEU and ROUGE when evaluating generative text.

Links
https://aws.amazon.com/compliance/programs/
https://owasp.org/www-project-top-10-for-large-language-model-applications/
https://aws.amazon.com/blogs/machine-learning/introducing-aws-ai-service-cards-a-new-resource-to-enhance-transparency-and-advance-responsible-ai/
https://aws.amazon.com/bedrock/pricing/
https://portkey.ai/blog/what-is-costar-prompt-engineering
https://www.reuters.com/legal/new-york-lawyers-sanctioned-using-fake-chatgpt-cases-legal-brief-2023-06-22/
https://aws.amazon.com/solutions/case-studies/doordash-bedrock-case-study/
https://zilliz.com/glossary/superglue
https://h2o.ai/wiki/squad/

https://huggingface.co/datasets/deepset/prompt-injections/viewer/default/train?p=1&views%5B%5D=train